Pioneer academics

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Pioneer Research Institute Evaluation Form

Scholar Information

Scholar's Full Name:	Vadlamani, Aikyan Satya
Scholar's Research Concentration:	The Random Walk of Mathematical Finance
Title of Scholar's Research Paper:	Using the Municipal-to-Treasury Ratio to Identify Relative Value Opportunities

Grading and Academic Oversight Section

Grade conferred by professor: A

**Full grading rubric approved by Pioneer Academics and Oberlin College*

The Pioneer Research Institute's college accreditation and academic oversight are conducted collaboratively with Oberlin College:

Listing on Common Application:	"Summer Program, Credit Awarded Directly by Oberlin College"
Course Title on Coalition Application:	"099" "Pioneer Research Course"
College Course ID on UC Application:	"INST 099" "Pioneer Research Course"

Comprehensive Evaluation Section

Evaluator Name & Full Title:	Hesam Oveys, Clinical Associate Professor of Mathematics, Courant Institute of Mathematical Sciences Assistant Director of Undergraduate Studies for Instruction
College/University & Department:	New York University, Courant Institute, Dept. of Mathematics
Full Educational Background of Evaluator:	Ph.D., Mathematics, University of Missouri (2015)

Please briefly explain the nature and requirements of the research paper and your interaction with the scholar:

The research concentration 'The Random Walk of Mathematical Finance' challenged scholars to author an original, rigorous research paper examining fixed-income relative value and spread dynamics. Topics included stochastic yield curve modeling, mean-reversion processes, Ornstein-Uhlenbeck dynamics, and cointegration testing. Scholars were required to retrieve historical yield series, construct statistical trading signals in Python, backtest systematic relative-value strategies, and analyze risk-adjusted performance. The final deliverable required synthesizing fixed income institutional mechanics with formal econometric backtesting.

Aikyan was an extraordinary scholar whose intellectual maturity and quantitative independence stood out from day one. In our group lecture sessions, he demonstrated a strong grasp of financial market structure, frequently asking insightful questions about bond yield spreads and tax-exempt liquidity. During our individual sessions, he executed an innovative fixed-income research strategy on the Municipal-to-Treasury (M/T) ratio with minimal supervision. He implemented my feedback well and displayed exceptional professionalism in all interactions.

Please rate the scholar in the following areas:

	Excellent	Good	Average	Below Average	Poor
Ability to form original ideas and concepts	X				
Ability to communicate thoughts in an effective and articulate manner	X				
Ability to synthesize and organize information from disparate sources	X				
Level of scholar's curiosity, aptitude, and industriousness	X				

Based on the scholar's performance throughout Pioneer, how would you rate this scholar's potential for undergraduate-level academic work at a top college/university, relative to other undergraduate students whose academic work you have mentored?

One of the best I have encountered, top 5%	**Excellent, top 10%** X
Very Good, top 25%	Good, top 50%
Below Average	

Describe some of the scholar's strengths:

Aikyan demonstrates an advanced grasp of fixed income mathematics and institutional market mechanics well beyond his years. His formulation of an Ornstein-Uhlenbeck mean-reverting framework to generate systematic trading bands on the 10-year and 30-year Muni/Treasury ratio is brilliant. His backtest accounted for real-world trading constraints, including transaction costs and tax-equivalent yield adjustments, yielding impressive risk-adjusted Sharpe ratios across multiple rate cycles.

Describe some areas in which the scholar can improve:

To build upon his strong fixed-income foundation, Aikyan could incorporate formal econometric stationarity and cointegration testing, such as Augmented Dickey-Fuller or Johansen tests, to mathematically validate mean-reverting dynamics across changing interest rate regimes. He might also explore multi-factor term structure models or examine how credit risk shocks during periods of municipal distress affect the spread. Continued study in continuous-time finance and stochastic calculus will help him further formalize the theoretical foundations of fixed-income arbitrage.

While the second half of the research work was one-to-one with the professor, the first half involved a small research cohort of 3-6 scholars. Describe the contribution that the scholar made to the research group discussion and learning:

Aikyan was a key intellectual contributor in our group seminars, consistently elevating the standard of discussion. During our initial discussions on trading mechanisms and market microstructure, he shared thoughtful insights into how institutional capital flows drive fixed income spreads. His articulate contributions helped enrich the lecture experience for all members of the cohort.

Would you recommend this scholar to a college admissions officer at a top college or university? Why or why not?

I give Aikyan my highest possible recommendation to a college admissions officer at a top college or university. He possesses deep analytical rigor, remarkable intellectual maturity, and the independent research capabilities of a seasoned university student. He is destined to be a standout scholar and will excel in the most challenging academic environments.

Evaluator's Signature:	
Evaluator's Printed Name:	Hesam Oveys
Date of Evaluation:	September 20, 2026
Evaluator's Email:	hoveys@cims.nyu.edu

Pioneer Academics affirms that the scholar whose research paper is attached has followed Pioneer Academics' protocols for developing original research and that their paper has met Pioneer Academics' standards for authenticity. The grading and evaluation have met the standards collaboratively defined by Pioneer Academics and Oberlin College & Conservatory. If there are any questions regarding the work's academic integrity, please contact us at academic.integrity@pioneeracademics.com or by calling (855) 572-8863. We will conduct an immediate review and respond to your questions promptly.